Two contact rays replace the probability panels at large indices

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The coarse acquisition need export only two numbers to the comparison at $n \geq 700$: a bound for nD and a bound for the raw threshold. Its later probability-panel calculations can be bypassed. The argument below acquires one probability interval and a common conditional amplitude $3/5$ using only scalar inequalities and two contact rays. It does not replace the Fourier calculation that produced the coarse bound.

Put $\sigma = \sqrt{10}$, $p_* = (4-\sigma)/2$, and $B_* = 32.20665378$. Every terminating decimal below is an exact rational. For a compactly supported centered variance-one law define

$$\rho = \mathbb{E}|X|^3, \quad \gamma = \mathbb{E}X^3, \quad \beta = \mathbb{E}[X|X|], \quad \tau(t) = \mathbb{E}[X\mathbf{1}_{X\geq t}],$$

$$J = \sum_{(a,b)} \tau_{a,b}(b-a)^2, \qquad D = (3+\sigma)\rho - \gamma - 3J.$$

The sum ranges over bounded open support gaps. There are at most countably many; all nonnegative gap sums are obtained by monotone convergence from finite subcollections.

Theorem 1 (Scalar post-entry acquisition)**.** *Suppose $n \geq 700$ and $nD \leq B_*$. Then zero is outside the closed support, and there is an actual central gap $(-A_-, A_+)$ with positive endpoints. For the actual sign $B = \mathbf{1}_{X>0}$ put*

$$p = \mathbb{P}(B=1), \quad q = 1-p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_-, \quad \ell = m/v, \quad X = \ell(B+U-p).$$

The conditional noises U_j are centered. With

$$\theta = \mathbb{E}\{X - \mathbb{E}(X \mid B)\}^2, \quad a_0 = p - A_-/\ell, \quad a_1 = q - A_+/\ell, \quad d = a_0 + a_1, \quad \lambda = n\mathbb{E}U^2 = n\theta/\ell^2,$$

one has

$$3/8 < p < 93/200, \qquad \theta < 9/2500, \qquad d < 1/250, \qquad \ell > 499/250, \tag{1}$$

$$\rho < 9/8, \qquad .066148 < \beta < .2527, \qquad \lambda < 7/20. \tag{2}$$

If $p \leq 2/5$, then additionally $\lambda < 8/25$. No contact, predecessor, exported probability interval or numerical support radius is a premise of these conclusions.

1 The actual gap and a short scalar localization

Throughout this proof $D < D_0 := .047$, since $700D_0 > B_*$. If zero belonged to the support, every gap would lie on one side of it and $(b-a)^2 \leq 2\int_a^b |t|\,dt$. Tonelli gives $J \leq \rho$, hence $D \geq \sigma\rho - \gamma \geq \sigma - 1 > D_0$, a contradiction. Centering and positive variance supply both signs and the stated gap. Conditional means give $a_0, a_1 \geq 0$ and $0 \leq d < 1$. Write $w = 1-\theta = v\ell^2$ and

$$Q(p) = (\sigma+1)p^2 + (\sigma-1)q^2 - 6pq = (6+2\sigma)(p-p_*)^2.$$

For the residual-noise measures $\int f\,d\mu_\pm = \mathbb{E}[(X-\mathbb{E}(X\mid B))^2\mathbf{1}_{\{\pm X>0\}}f(|X|)]$, the elementary outer-gap integral bound gives

$$D \ge \ell w Q + E_0 + \int k_+ d\mu_+ + \int k_- d\mu_-, \tag{3}$$

where

$$E_0/\ell = 6w\{d(1-d)+a_0a_1\}, \quad k_+(x) = (\sigma-1)(x+2\ell q)+6A_+,$$
$$k_-(x) = (\sigma+1)(x+2\ell p)+6A_-.$$

Indeed $J_o^+ \le 2\int_{A_+}^\infty (t-A_+)\tau(t)dt = \mathbb{E}[X_+(X_+-A_+)^2]$, and similarly on the negative side. Add $m(A_-+A_+)^2$ and substitute the conditional means. This is the common cost identity, before discarding any positive term. It applies to endpoint atoms and countably many gaps.

The weaker gap bound $J \le \rho + 2mA_-A_+$ gives

$$A_-A_+ \ge z := \frac{3.162277660 - 1 - D_0}{3} > 0, \qquad w \ge A_-A_+ \ge z.$$

Here $\rho \ge 1$, $\gamma \le \rho$, $m \le 1/2$, and $A_-A_+ \le (\ell p)(\ell q) = w$ were used. Since all terms in (3) are nonnegative,

$$D \ge w^{3/2}Q/\sqrt{v} \ge 2z^{3/2}Q.$$

The positive-base squared tests $4z^3Q_-(.36)^2 > D_0^2$ and $4z^3Q_-(.48)^2 > D_0^2$, where Q_- replaces σ by its lower rational bound, exclude $p \le .36$ and $p \ge .48$. This uses the monotonicity of the squared distance to $p_* \in (.36,.48)$, not an initial probability panel.

On $.36 < p < .48$, $\ell \ge 2\sqrt{z} > 1.67$ and

$$\frac{k_+}{\ell} \ge q\{(\sigma+5)z/w + 2(\sigma-1)\} > 5.2, \quad \frac{k_-}{\ell} \ge p\{(\sigma+7)z/w + 2(\sigma+1)\} > 5.2.$$

The first inequality uses $x \ge A_+$ and $A_+ \ge z/(\ell p)$; the second is its signwise counterpart. Since $w \le 1$, the asserted constants follow at $q = .52, p = .36$. The total mass of $\mu_+ + \mu_-$ is θ, so $D > 1.67(5.2)\theta$ unless $\theta = 0$. Consequently $\theta < .006$, $w > .994$, and $\ell > 1.99$. Also $A_- + A_+ \ge 2\sqrt{z}$ gives

$$d \le 1 - 2\sqrt{zv/w} < .2, \qquad v \ge .36(.64).$$

Dropping a_0a_1 and using the lower endpoints in the kernels gives

$$D \ge \ell\{wQ + 6wd(1-d) + \theta\min(C_+ - b_+d, C_- - b_-d)\}, \tag{4}$$

where $C_+ = 3(\sigma+1)q$, $C_- = 3(\sigma+3)p$, $b_+ = \sigma+5$, and $b_- = \sigma+7$. On the current domain both noise coefficients are positive, and each branch increases in d, since $6(.994)(.6) - .006(3.162277661 + 7) > 0$. The inward term alone excludes $d \ge .004$:

$$1.99(.994)6(.004)(.996) > .047.$$

Here $d(1-d)$ increases up to $1/2$, so no larger root was selected.

At $d = 0$, the derivative with respect to θ of $\sqrt{(1-\theta)/v}\{(1-\theta)Q + \theta C\}$ has the sign of $(2-3\theta)C - 3(1-\theta)Q$. It is positive because $\theta < .006$, $C > 6.49$, and $Q < .05$ on $[.36,.48]$. Thus (4) implies $D \ge Q/\sqrt{v}$. On $[.36, 3/8]$ this ratio decreases, and the squared endpoint test $Q_-(3/8)^2 > D_0^2(3/8)(5/8)$ excludes that interval. On $[93/200, .48]$, use $Q \ge Q(93/200) > D_0/2$ and $\sqrt{v} \le 1/2$. This proves the probability interval in (1); no incorrect monotonicity of $Q/\sqrt{v}$ below $1/2$ is needed on this right interval.

Now $C \ge \min\{3(\sigma+1)(107/200), 3(\sigma+3)(3/8)\}$. The bound $D \ge \ell\theta C$, with $\ell > 1.99$, gives $\theta < .0036$ and therefore $\ell > 1.996$. The exact identity

$$\rho = w^{3/2}\rho_B(p) + \int (x+2\ell q)d\mu_+ + \int (x+2\ell p)d\mu_-, \quad \rho_B(p) = \frac{1-2v}{\sqrt{v}},$$

and $k_+ \ge (\sigma-1)(x+2\ell q)$, $k_- \ge (\sigma-1)(x+2\ell p)$ show

$$\rho \le \rho_B(p) + D/(\sigma-1) < 1.1 + .047/(54/25) < 9/8.$$

Here $v \ge 15/64$ and $\rho_B < 1.1$ follows by a rational square test. The exact signed-second-moment identity gives $1-2p-2q\theta \le \beta \le 1-2p+2p\theta$. Its two endpoint evaluations give (2).

2 Noise from the same marginal budget

The noise estimate uses the one-sided cubic inequality $\mathbb{E}Z^3 \le at - (t^2/a)$ for a centered $Z \le a$, $t = \mathbb{E}Z^2$. It follows by expanding $\mathbb{E}[(a-Z)(Z+t/a)^2] \ge 0$. When $a = 0$, centering forces zero noise and its term is absent. For $\lambda_0 = nq\mathbb{E}U_0^2$, $\lambda_1 = np\mathbb{E}U_1^2$ and any $\bar{d} \ge d$, the common budget yields

$$nD \ge \ell^3\{nvQ + \lambda_0 Y_0(p) + \lambda_1 Y_1(p)\}, \tag{5}$$

$$Y_0 = 3(\sigma+3)p - (\sigma+7)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma+1)p},$$
$$Y_1 = 3(\sigma+1)q - (\sigma+5)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma-1)q}.$$

For clarity, before arithmetic–geometric mean the extra terms are $6nv\{d(1-d)+a_0a_1\}$ and $(\sigma+1)\lambda_0^2/(nqa_0)+(\sigma-1)\lambda_1^2/(npa_1)$. Use $d(1-d)+a_0a_1 \ge (1-\bar{d})(a_0+a_1)$ and pair each summand with its corresponding reciprocal term. This proves (5) without an independent allocation of the conditional variances; it is the general v13 inequality, not its old numerical table.

Take $\bar{d} = .004$. Both Y_0 and Y_1 increase in their respective arguments p and q. Exact root comparisons at $p = 3/8, q = 107/200$ give $Y_0 > 12.8$ and $Y_1 > 11.8$: lower bounds 3 and 2.6 suffice for the two square roots. On $p \le .4$, use $q \ge .6$ and square-root lower bound 2.7 to get $Y_1 > 12.8$. For $\lambda > 0$, since $\lambda_0 + \lambda_1 = \lambda$,

$$B_* \ge nD > 1.996^3(11.8)\lambda, \qquad \lambda < .35; \quad p \le .4 \Longrightarrow B_* \ge nD > 1.996^3(12.8)\lambda, \quad \lambda < .32.$$

At zero noise the conclusions are immediate. This finishes Theorem 1.

3 One pair of contact rays

Theorem 2 (A common conditional strip)**.** *Under Theorem 1, suppose also that (X,H) is an attained fixed-target pair at C_E with nonnegative excess and true sharp predecessor $C_{n-1} \le C_E$, so that the retained closed-support contact/heat bounds below apply. Then*

$$-.6 < U_0 \le a_0, \qquad -a_1 \le U_1 < .6.$$

No bound on H is needed for this support conclusion.

Put $S = 81/80$, $\eta = 8/875$, and $c_0 = 819/2000 < C_E$. For $F_v(x) = \Phi(x/\sqrt{v})$ the retained contact function is

$$\frac{H_n(H,t)}{n^{3/2}} = F_{n-1}(H-t) - F_n(H) + tF_n'(H) - \frac{t^2-1}{2}F_n''(H).$$

Closed-support contact gives $0 \le H_n(H,\pm t) + C_E Q(t;r,b)$ after discarding the nonpositive excess term. The all-threshold heat lemma gives, for $t > \sqrt{3}$,

$$H_n(H,\pm t) \le h(t) := S^3\left\{\frac{2}{5}\left(\frac{t^3-3t}{6} + \frac{|t^4-6t^2+3|}{72}\right) + \frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2-2)}{78720}\right\},$$

$$|\partial_t H_n(H, \pm t)| < g(t) := \frac{2}{5}(t^2/2 + 1/2 + \eta).$$

At a support contact the upper expression is $H_n(H, \pm t) + C_E Q(t; r, b)$, where $Q(t; r, b) = -t^3 + 3rt^2/2 + 3bt + \eta r$. Here $r = 9/8$, with $b = -.066148$ on the negative ray and $b = .2527$ on the positive ray. These upper replacements are sign-correct.

Use just the two rational boundaries

$$t_- = 1.996(3/8 + .6), \qquad t_+ = 1.996(107/200 + .6).$$

At each, the exact checker proves the eight strict inequalities

$$t^2 > 3, \quad h + c_0 Q < 0, \quad Q, Q', Q'' < 0, \qquad L, L', L'' < 0, \quad L = g + c_0 Q'.$$

The boundary upper bounds are respectively less than $-.40$ and $-.13$. Since $Q''' = -6$ and $L'' = 2/5 - 6c_0 < 0$, all continuation signs persist along the entire ray. The derivative of the actual upper function is strictly below L, because $Q' < 0$ and $C_E > c_0$. Thus both exterior rays, including boundary atoms, contain no support point. This differentiates the exact heat expression, not its absolute-value majorant.

Finally $X > -t_-$ implies $U_0 = X/\ell + p > -(3/8 + .6) + p > -.6$. Similarly $X < t_+$ implies $U_1 = X/\ell + p - 1 < 107/200 + .6 + p - 1 < .6$. The inner endpoints and conditional centering are the actual ones already defined. This proves Theorem 2.

The resulting short chain, and its limits

The selected coarse-entry theorem exports $nD \le B_*$ and $|H| \le 1.83605680 < 1.84$. The two theorems above then supply, on one interval $(.375, .465)$, the support and noise premises for the v15 local unions: use the left union when $p \le .4$ and the right union otherwise. Their strict signed conclusions exclude an acquired nonnegative witness at every $n \ge 700$.

No v10 numerical localization, one-pass panel export, v11 left-strip test, v13 degree-eight noise table or low/high-probability exclusion is needed on this route. The underlying common-cost and one-sided-cubic lemmas remain and have been displayed here. The manuscript's two coarse passes, four signed passes and complete Fourier inputs remain ancestors of B_* and the raw-threshold ceiling. Their internal panels and intermediate support gaps cannot be erased merely because these outputs are enough downstream. The strict predecessor, attainment, reflection, and finite base are still separate obligations. In particular this is not an unconditional cutoff at 700.

The companion audit gives source locations and a dependency graph. Its arithmetic script uses exact rationals and exact $\mathbb{Q}(\sqrt{10})$ actual-law checks. A rare conditional atom at $-.8$ satisfies the geometric budget and even the old radius 3.14, but violates the .6 strip; it explicitly shows why contact cannot be dropped.